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Abstract

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Preface

Preface.

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Abbreviations

API	Application programming interface
BLR	Bayesian linear regression
CDW	Construction and demolition waste
CE	Circular economy
GBC	Green Building Council
GHG	Greenhouse gas
MVG	Multivariate Gaussian

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1 Introduction

2 Background

This chapter establishes the reasoning behind the urgent need for a global transition towards a circular economy (CE) and describes the role digitalization plays in the process. It takes the Finnish construction industry as an example of this transition, and specifies the key challenges and opportunities of driving the change with data and digital solutions.

2.1 Circular economy and digitalization

It is no news that the ongoing environmental crises, such as global warming and loss of biodiversity, pose a large part of the most significant threats of our lifetime. The challenge of steering humankind towards a sustainable way of life is monumental, and requires extraordinary feats, as well as cross-disciplinary coordination and co-operation from all entities ranging from individual people to businesses, industries, countries and international alliances.

In 2020, the Finnish Innovation Fund Sitra conducted a study on the megatrends for the beginning decade [4], identifying five different, but intertwined global megatrends, three of which are the following:

- *The urgent need for ecological reconstruction*
- *The redefinition of the economy*
- *Technology is embedded in everything*

Getting the interactions between technology and economy to work in favour of fighting environmental crises is an incredibly complex, yet critical challenge. In other words, it is safe to state that the two latter trends play a key role in the way we drive ecological reconstruction in the years to come.

The culprits behind our environmental crises are manifold, but at the end of the day, much of the problem boils down to how we use the natural resources our planet provides us with. Already in 2019, resource extraction and processing made up approximately half of global greenhouse gas (GHG) emissions, in addition to more than 90% of biodiversity loss and water stress. [13] It has become evident, that the sun has set on the old linear economy model. We can not afford to extract, use and dispose of resources with the pace we have reached due to assumptions of limitless economic growth with little to no regard to external costs. The concept of a circular economy presents an alternative model to this, where product lifetime and reusability is maximized, production waste is minimized, and business models based on sharing and reusing products are implemented. [8] The potential in these strategies is enormous: according to a Material Economics study from 2018 [6], the transition to a more circular economy could reduce GHG emissions from heavy industry in the EU by 56% by 2050. Contrary to a common misconception, this shift does not need to happen at the cost of economic growth as a whole. Aligning technologies and business models to tackle wasteful value creation chains and shift towards a more

circular model would actually create new markets, and boost employment, lifting EU's GDP by 7% and yielding annual benefits of up to €1,8 trillion by 2030. [7]

As Sitra states, technology is embedded in everything, and the transition to CE is, of course, no exception. The breathtaking pace digitalization has developed with during the last decades has presented us with countless possibilities and opportunities, but also with its own set of challenges. For example, with increasing amounts of digital solutions and services increases also the demand for energy. Hence, it would be naïve to assume digitalization to be a "silver bullet" that directly solves the problems related to implementing CE models, as stated in a 2019 study by the European Policy Centre (EPC). [8] That being said, the study focuses on establishing the role of digitalization as a key enabler of CE - as long as it is properly managed. They divide approaches to use digitalization to accelerate the transition to CE into three categories:

1. Improving knowledge, connections and information sharing
2. Making business models, products and processes more circular
3. Strengthening the roles of citizens and consumers

Within the scope of this thesis, we will mainly focus on categories 1 and 2, exploring the ways proper data aggregation, utilization and sharing can support businesses in adapting more circular models. Specifically, we will look at the construction industry, as it is responsible for almost half of global raw material extraction and a quarter of global GHG emissions. [5]

2.2 Circularity in the Finnish construction industry

When reviewing CO₂ emissions as part of the environmental impact of the construction industry, it is essential to divide them into three categories according to the phases in the lifecycle of a building:

1. **Upfront carbon** (materials production and construction phase)
2. **Operational carbon** (use phase, energy consumption)
3. **End of life carbon** (demolition phase, demolition waste handling)

These are defined by the World Green Building Council (WorldGBC) in their 2019 report "Bringing embodied carbon upfront" [1]. The same study finds that with the recent advances in the field of renewable energy, operational carbon emissions are reduced significantly. Consequently, upfront and end of life carbon will grow quickly in proportion. In particular, the WorldGBC predicts that upfront carbon will be responsible for up to half of the whole carbon footprint of new construction between the the present and 2050. A similar trend can be seen in Finland, as well. Up until 2020, approximately 80% of the Finnish built environment's total lifecycle emissions came from operational carbon. [10] With low-carbon energy gaining foothold in the Finnish energy market, focus is shifting increasingly towards reducing upfront and

end of life carbon by transitioning to a more circular operating model. The Finnish government has recently taken the lead in driving this transition across industries in Finland. In the spring of 2021, they established a strategic programme [14] to pave the way towards a new Finnish economy based on CE by 2035. The programme dedicates a significant amount of focus on the construction industry, and specifies several action points to promote CE in its context, including financial incentives, digitalization and data management.

From a CE point of view, one of the the main challenges in the Finnish construction industry is construction and demolition waste (CDW) management. Legislation and various costs seem to form a setting, where there is no clear and desirable way for the responsible party to handle CDW. On a demolition site, for example, the prime contractor owns all of the materials and waste (although the responsibility of demolition waste handling lies with the demolition contractor, if the demolition is outsourced by the prime contractor), and therefore is also responsible for processing all of the material streams released during demolition accordingly, including all related costs and taxes. Current legislation makes it difficult, and in many cases expensive, to arbitrarily dispose of CDW. Obviously, this option is also undesirable from an ecological point of view, due to which it is also safe to assume that legislation is going to get even more strict in this matter. Enormous costs and the lack of proper real estate, on the other hand, make it impractical to store CDW even for short spans of time. These matters lead to the conclusion that there is also a significant cost case in addition to the ecological case, that speaks for a circular system where CDW could be salvaged immediately from site for reuse (some components can be extracted pre-demolition and used again as such) and recycling (waste that is not feasible for reuse is processed into new products).

There are, however, significant challenges to overcome. Plenty of costs are embedded in the process of preparing CDW from extraction to reuse and recycling, and said costs often outweigh the costs of disposing of this waste, which makes it less appealing for contractors to salvage it. This problem tends to have the effect of making virgin materials more attractive price-wise, which hinders the demand for reusable and recycled CDW, and makes adapting a more circular model difficult from the point of view of procurement as well. Not only that, but the whole paradigm of construction and most of its processes are built around using virgin materials; a buyer generally risks lower product quality when dealing with reused and recycled materials. This problematic is clearly a key obstacle, as it effects negatively both the supply and demand sides of the non-virgin construction material market. This has also been recognized by the Finnish government, as it has recently set in motion investigation projects to e.g. identify ambiguities in construction product qualification procedures. [19] As an example result, the Finnish Ministry of the Environment released a policy brief [20] on verifying the feasibility of reused materials in June 2022. It states that it is sufficient to check for feasibility with construction site -specific verification, which means that a CE-marking is not required for e.g. demolition waste that is reused as such.

However, even if there was a strong financial incentive to favor non-virgin materials, a significant challenge remains aligning demand with supply in the reusable and

recycled CDW market on a temporal level. Currently, contractors are able to release information of available materials only as soon as they become available, as the demolition project proceeds. Because storage possibilities for these materials are few and costly, the time window to find a buyer is far too narrow. Additionally, this unpredictable batch-like supply leads to discontinuous availability, rather than a continuous supply of non-virgin materials. To counter this challenge, Finnish officials are working on a series of databases and registers that would together provide a centralized data storage system for information on demolition projects and plans, construction product characteristics and material streams, to name a few. These systems are planned to come with diverse APIs to enable private entities to develop their own services and platforms to utilize all of the data stored in these public systems. The following section will introduce one such project, in which the author is also involved through his employment.

2.3 Motiva’s data platform and ecosystem for material circulations

In order to make the raw data from the existing and upcoming registers for construction materials and demolition waste available and usable to companies and consumers, it is vital that different services are developed to refine and process the data. Even though work on the state-wide registers is still in progress, the Finnish state-owned company Motiva Ltd. is already working on a platform for construction material circulation. Motiva is a company of experts specialized in providing organizations and consumers with information, solutions and services on how to use energy and materials more efficiently and sustainably. With this platform, it is Motiva’s aim to build a hub that collects information from various sources under a single platform, acts as a meeting place for supply and demand of non-virgin construction materials and supports all relevant stakeholders in adapting to a more circular operating model in the construction industry.

At the moment, this platform is still in the planning phase. Motiva has developed the concept for it, and it has been validated and further developed using two different projects as case studies. The first one is a demolition project that takes place in Vattuniemi, Helsinki, where 16 buildings are set to be demolished during 2022-23. The other site is located in the Tampere area, where especially the reuse and circulation of ashes as a construction product in infrastructure building was explored. The latter project ended in Spring 2022 for the time being. These projects are vital in gathering field experience on the challenges and requirements of material circulation. Specifically, Motiva seeks to pilot planned features of their platform, and to gain knowledge on how exactly predictive, data-driven demolition planning supports a circular economy.

2.4 The potential of pre-demolition audits

A critical part of shifting towards predictive demolition planning are the procedures that guide the mapping of the demolition site. These include mapping of harmful,

but also reusable and recyclable materials that the building to be demolished consists of. Historically, only the mapping of asbestos and other harmful materials has been required by law, but nowadays it is increasingly common to do a broader material mapping. The most extensive mapping process is the pre-demolition audit (in Finnish "purkukartoitus"), which also includes, for example, reusable movables and recommendations on how to reuse, recycle or dispose of the material streams released during demolition. [11] These audits are at the moment voluntary, but officials are constantly planning ways to encourage real estate owners and contractors to use them as a tool to increase material circulation. This is slightly challenging, since a more extensive audit is also more expensive to the client. It is, however, crucial from the point of view of CE, that pre-demolition audits are carried out as carefully as possible, because gaining knowledge of future material streams is necessary in order to align demand with supply in the non-virgin material market.

Carrying out pre-demolition audits, and especially estimating the amounts of different materials released, is hard to do accurately. All experts interviewed for this thesis agreed on this, and stated several possible reasons for the significant errors that occur even in state-of-the-art audits. Some of the material amounts are especially hard to predict, because they might often be hidden from plain sight in the foundations of the building in question. An example of this kind of material is mineral wool, which is often used as an insulator. Especially in the case of old buildings, where the quality of material documentation might be suboptimal, these kind of "hidden" materials can be nigh unto impossible to predict. Extensive, on-site analyses might yield better results, but these are mostly impractical, because the target buildings are often still in use at the time of conducting the audit. It might also happen, that some renovations or other changes during the building's lifetime have not been properly documented, which causes some material streams to come as a surprise during demolition. Finally, it is also natural that clients want to minimize the costs of a demolition project, and the pre-demolition auditors may have to compromise extensiveness and accuracy of the audit in order to stay within the budget.

Given the potential of pre-demolition audits from a CE point of view, and the nature of the challenges described above, it is evident that making the audit process more data-driven holds significant possibilities to develop construction material circulation. The obvious bottleneck at the moment, however, is the lack of data. There are no standardized, mandatory procedures to store pre-demolition audit, material mapping, or realized material stream data. This leads to the relatively small amount of feasible data to be scattered all over different stakeholders, and to have little to no quality guarantees. This challenge is at least partly being addressed by the projects described in sections 2.2 and 2.3, but this thesis will motivate and set a roadmap for proper data aggregation, utilization and sharing in this industry by demonstrating predictive capabilities using actual proof-of-concept models.

3 The Models

This chapter introduces two distinct modeling approaches, the Random Forest regressor and Bayesian Linear Regression (BLR), to predict demolition material streams using initial knowledge of the target building. It is imperative to note, that these models are only a proof of concept - they are meant to demonstrate that given proper data, it is possible to use various machine learning algorithms to make feasible predictions on different material streams, and thus support the pre-demolition audit process. Hence, these models as such should not be relied on in actual cases.

3.1 On notation

Before applying the models on actual data in section 3.4, we will consider a general supervised learning setting with a dataset $\mathcal{D} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_m\}$, where a vector $\mathbf{z}_i$ denotes a data point consisting of features and labels, and m is the size of the dataset. The feature space $\mathcal{X}$ encloses all possible feature values, and similarly, the label space $\mathcal{Y}$ is the space of all possible label values. In the case of $\mathcal{Y}$ being discrete, we talk about classification problems. In the problem setting of this thesis however, we have $\mathcal{Y} = \mathbb{R}$, which implies a regression problem.

Let n be the number of features in the dataset. The feature vector $\mathbf{x} = [x_1, x_2, \dots, x_n]$ contains all feature values for a single data point. It is often useful to represent all feature vectors in a $m \times n$ matrix

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_m \end{bmatrix}.$$

The label of the i th data point is denoted by y_i , and by extension, all labels in the dataset are represented in the label vector $\mathbf{y} = [y_1, y_2, \dots, y_m]^T$. In the scope of this thesis, we will only consider a single label at a time in order to avoid higher dimensional label matrices.

3.2 Random Forest regression

Introduced by Breiman (2001) [3], the Random Forest algorithm is a supervised ensemble learning method, that is made up by of a group of decision trees. Decision trees and random forests have remained a popular modeling approach because they imitate human reasoning and decision making, and are thus conceptually intuitive. They can be used for classification and regression tasks alike, but we will focus on the latter because of the nature of our problem setting.

3.2.1 Decision trees

A Decision Tree regressor is a predictor that maps features to the label space by essentially going through a series of if-else-statements. Each of these decision

statements checks, whether or not a specific subset of features fills the decision criteria. This is done until the process arrives at a prediction. It is conventional to represent decision trees as graphs, that start at the top with a (root) decision node, and continue down to further decision nodes, until arriving at a leaf node. Here the decision nodes represent the if-else-clauses, and leaf nodes the predictions. A simple example of such a decision tree for three-dimensional data $\mathbf{z}_i = [x_1^{(i)}, x_2^{(i)}, y^{(i)}]$ is depicted in Figure 1. This tree shatters a training dataset into four subsets, one for

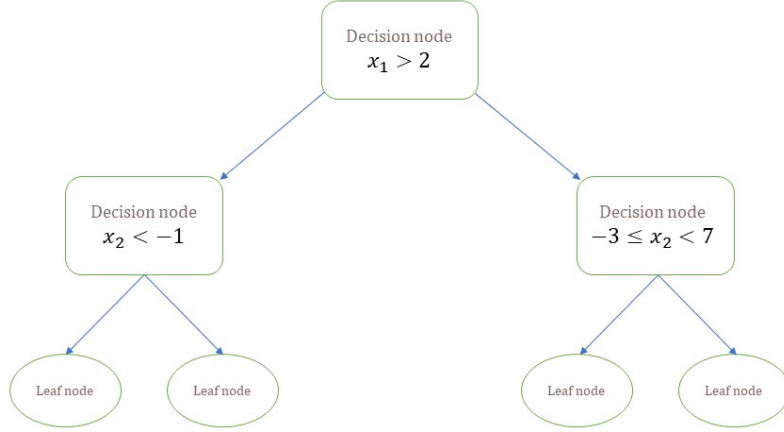


Figure 1: A graph representation of a simple decision tree.

each leaf node. It achieves this by taking one data point at a time, starting at the root node, and choosing the next edge to traverse based on whether the decision statement of the node is true or false in the case of the feature vector in question. If the chosen edge leads to a leaf node, the tree starts to process the next data point. If the next node is a decision node, the decision process is repeated. A new prediction is made similarly by passing the new feature vector through the decision nodes, until it lands on a leaf node. The predicted label is then given a value equal to, for example, the mean of the training labels in the subset associated with that leaf node.

An obvious point of interest remains the question of how to grow an optimal decision tree. There are several possibilities to do this, but we will consider a variant of the ID3 (Iterative Dichotomizer 3) algorithm [15] (Algorithm 1), designed to handle regression problems. The algorithm starts with a training set $\mathcal{D}$ and a root node. It then proceeds to recursively build the two subtrees sprouting from the root node, until maximum tree depth k is reached, or the size of the resulting subset is small enough ($< l$).

The final question to be cleared is the problem of finding the best possible splitting rule, or decision statement, at each decision node. Generally, the metric for the "goodness" of a split is called the *Gain*. Again, there are multiple possibilities when it comes to choosing a Gain function [17], but our choice will be *variance reduction*.

Let $\mathcal{D}_0$ be the dataset at the current node, and $\mathcal{D}_i, \mathcal{D}_j$ be its split, such that $\mathcal{D}_i \cap \mathcal{D}_j = \emptyset$ and $\mathcal{D}_i \cup \mathcal{D}_j = \mathcal{D}_0$. With $\mathcal{Y}$ denoting the set of labels, we can define

the (weighted) variance reduction as

$$\Delta\sigma^2 = \text{Var}(\mathcal{Y}_0) - \left(\frac{|\mathcal{D}_i|}{|\mathcal{D}_0|} \text{Var}(\mathcal{Y}_i) + \frac{|\mathcal{D}_j|}{|\mathcal{D}_0|} \text{Var}(\mathcal{Y}_j) \right). \quad (1)$$

The algorithm goes through every possible split and finally chooses

$$\mathcal{D}_1, \mathcal{D}_2 = \arg \max_{\mathcal{D}_i, \mathcal{D}_j \subseteq \mathcal{D}_0} \Delta\sigma^2$$

to be the respective subsets for the next nodes.

It is worth to mention, that letting the tree grow arbitrarily large results in an infinitely large Vapnik-Chervonenkis dimension for the model, and consequently in unacceptably severe overfitting. [17] Therefore, it is critical to set the bounds k, l

Algorithm 1 ID3 variant for regression

```

1: max_depth  $\leftarrow k$ 
2: min_samples  $\leftarrow l$ 
3: procedure ID3( $\mathcal{D}$ , current_depth = 0)
4:   nr_samples  $\leftarrow |\mathcal{D}|$ 
5:   if current_depth < max_depth & nr_samples > min_samples then
6:      $\mathcal{D}_1, \mathcal{D}_2 \leftarrow \arg \max_{\mathcal{D}_i, \mathcal{D}_j} \text{Gain}(\mathcal{D}, \mathcal{D}_i, \mathcal{D}_j)$ 
7:      $T_L \leftarrow \text{ID3}(\mathcal{D}_1, \text{current\_depth} + 1)$ 
8:      $T_R \leftarrow \text{ID3}(\mathcal{D}_2, \text{current\_depth} + 1)$ 
9:     return Decision node with left and right subtrees  $T_L$  and  $T_R$ 
10:  else
11:    return Leaf node with set  $\mathcal{D}$ 
12:  end if
13: end procedure

```

for tree depth and minimum split size. Optimizing these hyperparameters with more sophisticated methods than simple grid-search is out of the scope of this thesis, though.

3.2.2 Forming the ensemble

Another way to control overfitting and increase robustness of the Decision Tree model is to grow an ensemble of them and choose the average of all decision tree predictions as the final prediction. This method is called the Random Forest. In order to avoid correlated decision trees in the ensemble, the Random Forest algorithm trains each tree using a method called Bagging. [9]

The basic concept is to choose the training set for each tree by uniformly sampling a random subset with replacement from the original dataset. Additionally, the Random forest forces each tree to always consider only a random subset of features when choosing the best split at every decision node (line 6 of Algorithm 1). Introducing these random factors ensures that the generalization error of a random forest converges

as the number of trees in the forest increases [3], which makes it a strong tool to control overfitting.

At this point, the main weakness of the Random Forest becomes evident. Computational complexity quickly becomes a problem, especially as the number of trees increases. Growing a single decision tree, especially with a large training set, is a computationally heavy task due to the need to go through all possible splits at each decision node. This effect is amplified when growing a random forest, resulting in a theoretical computational complexity of $\mathcal{O}(k \cdot t)$ (excluding unification costs) and memory space cost of $\mathcal{O}(2^k)$, where k is the tree depth and t is the number of trees in the forest. [18]

3.3 Bayesian Linear Regression

The second, alternative modeling approach is Bayesian Linear Regression. It is arguably a slightly more advanced method, and offers the means to include domain knowledge in the model in the form of prior distributions, in addition to some metrics for uncertainty in the predictions. Given the scenario this model is applied to in this thesis, one could argue for BLR to be somewhat "overkill" of a choice. However, it goes to demonstrate that also more advanced probabilistic methods are applicable in this context.

3.3.1 A Bayesian approach to ordinary linear regression

We start with the familiar model of ordinary linear regression

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}, \quad (2)$$

where $\boldsymbol{\epsilon}$ is an error vector. Fitting this model is done by solving the optimization problem of finding the weight vector $\hat{\mathbf{w}}$ that minimizes the sum of squared errors (the so-called Ordinary Least Squares problem)

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathbb{R}^n} \sum_{i=1}^m |y_i - \mathbf{X}\mathbf{w}|^2 = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (3)$$

The Bayesian approach to this model follows the idea that we do not deal with exact values for the vectors $\mathbf{w}$ and $\boldsymbol{\epsilon}$, but rather place some probability distributions on them. Furthermore, we place prior distributions on the parameters of the afore mentioned distributions, which we then update with our data. The updated distribution is called *the posterior*, and it is computed with Bayes' rule

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}, \quad (4)$$

where $p(A|B)$ is the posterior (the probability of parameter A given the data B), $p(B|A)$ the likelihood (the probability of observing the data B given parameter A), and $p(A)$ is the prior on parameter A . The quantity $p(B)$ is another prior-like

marginal probability, but as we will see in section 3.3.2, it can be dealt with as a normalizing constant when deriving the posterior distribution.

Obtaining the posterior can be cumbersome; at times there might not even be an analytical solution available, and one has to resort to e.g. numerical integration. There is, however, the possibility of using a conjugate prior, which makes computations more straightforward and guarantees the existence of a closed-form posterior. [16] This essentially means, that using a prior that is conjugate to the likelihood results in a posterior of the same family of distributions as the prior.

Going back to the linear model in Equation (2), we now make the assumption that our data is given rise to by a multivariate normal distribution, or Gaussian (MVG). Knowing that the conjugate prior for a MVG likelihood is also a MVG [12], we place on the weights and errors, or noise, the prior distributions

$$\mathbf{w} \sim \mathcal{N}(\boldsymbol{\mu}_0, \alpha^{-1}\mathbf{I}) \quad (5)$$

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \beta^{-1}\mathbf{I}), \quad (6)$$

with α, β being hyperparameters interpreted as weight and noise precision, respectively. We can now write the likelihood as

$$p(\mathbf{y}|\mathbf{w}, \beta) = \mathcal{N}(\mathbf{X}\mathbf{w}, \beta^{-1}\mathbf{I}). \quad (7)$$

3.3.2 Deriving the posterior distribution

Arguably the most challenging part of BLR is coming up with the posterior for $\mathbf{w}$. Even with conjugacy, it still depends on the exact form of the priors and likelihood, and thus has to be derived carefully. Despite the lack of a general formula, we can still usually follow a certain routine to complete the derivation:

1. Start with Bayes' rule (4) and drop all normalizing constant terms independent of $\mathbf{w}$
2. Perform some algebraic manipulations
3. Due to conjugacy, recognize an unnormalized form of a distribution belonging to the same family as the prior

Following these steps, we can now derive the posterior for our case.

Using Bayes' rule (4) and our pre-defined distributions (5)-(7), we can express the (unnormalized) posterior as

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}, \alpha, \beta) \propto p(\mathbf{y}|\mathbf{w}, \beta)p(\mathbf{w}|\alpha) = \mathcal{N}(\mathbf{y}|\mathbf{X}\mathbf{w}, \beta^{-1}\mathbf{I})\mathcal{N}(\mathbf{w}|\boldsymbol{\mu}_0, \alpha^{-1}\mathbf{I}). \quad (8)$$

Writing out the two Gaussians, dropping constant terms and rearranging, we can

state that the product in (8) is proportional to

$$\begin{aligned}
& \exp \left[-\frac{1}{2} \beta (\mathbf{y} - \mathbf{X}\mathbf{w})^T (\mathbf{y} - \mathbf{X}\mathbf{w}) \right] \exp \left[-\frac{1}{2} \alpha (\mathbf{w} - \boldsymbol{\mu}_0)^T (\mathbf{w} - \boldsymbol{\mu}_0) \right] \\
& \propto \exp \left[-\frac{1}{2} \beta \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} + \beta \mathbf{y}^T \mathbf{X} \mathbf{w} - \frac{1}{2} \alpha \mathbf{w}^T \mathbf{w} + \alpha \mathbf{w}^T \boldsymbol{\mu}_0 \right] \\
& = \exp \left[-\frac{1}{2} \mathbf{w}^T \underbrace{(\beta \mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})}_{\boldsymbol{\Sigma}^{-1}} \mathbf{w} + \underbrace{(\beta \mathbf{y}^T \mathbf{X} + \alpha \boldsymbol{\mu}_0^T)}_{\boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1}} \mathbf{w} \right].
\end{aligned}$$

Completing the square [2], we get that the above expression is proportional to

$$\exp \left[-\frac{1}{2} (\mathbf{w} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{w} - \boldsymbol{\mu}) \right],$$

which we recognize as the unnormalized Gaussian $\mathcal{N}(\mathbf{w}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \boldsymbol{\Sigma} (\beta \mathbf{X}^T \mathbf{y} + \alpha \boldsymbol{\mu}_0) \tag{9}$$

$$\boldsymbol{\Sigma} = (\beta \mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})^{-1}. \tag{10}$$

This concludes the derivation, as we are left with a MVG with mean (9) and covariance (10) as the posterior distribution for the regression weights $\mathbf{w}$.

3.4 Data

3.5 Implementation

4 Results

4.1 Methods and results of model testing

5 Discussion

5.1 Meaning of modeling results

5.2 Roadmap for data aggregation and preservation

5.3 Future modeling possibilities

6 Summary

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